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Inventor(s)

Hansen, III; William S.

BAR LOCK ASSEMBLY

Abstract

A bar lock assembly includes a keeper having a base and a pair of laterally spaced side walls extending transversely from the base. At least one shaft extends between and is coupled to the side walls. A cam is coupled to and rotatable relative to the keeper between an engaged position and a disengaged position. The cam includes an engagement member engaged with the shaft when rotated to the engaged position. The engagement member is disengaged with the first shaft when rotated to the disengaged position. The cam may include a pair of engagement members. The keeper and/or cam may be roll formed and configured with a U-shaped cross-section.

Inventors: Hansen, III; William S. (Mettawa, IL)

Applicant: A.L. Hansen Manufacturing Co. (Waukegan, IL)

Family ID: 94823929

Assignee: A.L. Hansen Manufacturing Co. (Waukegan, IL)

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Background/Summary

RELATED APPLICATION [0001] This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/559,594, filed on Feb. 29, 2024, pending, the entirety of which is herein incorporated by reference.

TECHNICAL FIELD

[0002] The present invention relates generally to a bar lock assembly, and in particular to a bar lock having a keeper and cam, and also to methods of assembling and use the bar lock assembly.

BACKGROUND

[0003] Bar lock assemblies may be used for securing various components, such as trailer doors. The bar lock assemblies may include a keeper, a cam, an actuator shaft and a handle connected to the shaft. The handle may be manipulated to rotate the shaft and cam, thereby engaging the keeper and securing the component. Typically, the handle is configured to receive a lock to secure the bar lock assembly and component. The keeper and/or cam typically are made as die-cast parts, which are heavy, and relatively costly to manufacture and transport. Such parts also are not easily modified or capable of accommodating a locking device, such as a padlock. In addition, the handle, due an elevated positioning, may not be easily reached for receiving and manipulating the lock to secure the handle.

SUMMARY

[0004] Briefly stated, in one aspect, one embodiment of a bar lock assembly includes a keeper having a base and a pair of laterally spaced side walls extending transversely from the base, wherein the side walls are configured with laterally aligned openings. First and second longitudinally spaced shafts extend between and are connected to the side walls on opposite sides of the openings. An actuator shaft is rotatable about a rotation axis aligned with the openings. A cam is coupled to and rotatable with the actuator shaft about the rotation axis between an engaged position and a disengaged position. The cam includes an engagement member engaged with the first shaft when rotated to the engaged position. The engagement member is disengaged with the first shaft when rotated to the disengaged position.

[0005] In another aspect, one embodiment of a bar lock assembly includes a U-shaped keeper having a base and a pair of laterally spaced side walls extending transversely from the base. A shaft extends between and is connected to the side walls. A U-shaped cam includes a base and a pair of laterally spaced side walls extending transversely from the base, wherein the U-shaped cam opens toward and is nested between the side walls of the U-shaped keeper. Each side wall of the cam includes an engagement member. The cam is rotatable about a rotation axis between an engaged position wherein the engagement members are engaged with the shaft and a disengaged position wherein the engagement members are disengaged from the shaft.

[0006] In other aspects, various embodiments of a method of manufacturing the bar lock assembly includes roll-forming a keeper from a piece of flat sheet metal, and/or roll forming a cam from a piece of flat sheet metal.

[0007] The various aspects and embodiments provide significant advantages over other bar lock assemblies, and methods for the use thereof. For example, and without limitation, the keeper and cam may be easily manufactured and assembled, and are lighter in weight than conventional component, without sacrificing strength or durability. In addition, the components, such as the cam, may be easily modified to accommodate a lock device, thereby providing for an alternative and more accessible system for securing the bar lock assembly and component, such as trailer door.

[0008] The present embodiments of the invention, together with further objects and advantages,

will be best understood by reference to the following detailed description taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0009] FIG. **1** is a rear view of a trailer having a door and a bar lock assembly.
- [0010] FIG. **2** is a partially exploded perspective view of one embodiment of a bar lock assembly.
- [0011] FIG. **3** is a plan view of one embodiment of a bar lock assembly.
- [0012] FIG. **4** is a cross sectional view of the bar lock assembly taken along line **4-4** in FIG. **3**.
- [0013] FIG. **5** is an end view of the bar lock assembly shown in FIG. **3**.
- [0014] FIG. **6** is a flat pattern for one embodiment of a keeper.
- [0015] FIG. **7** is a perspective view of one embodiment of a keeper.
- [0016] FIG. **8** is a side view of the keeper shown in FIG. **7**.
- [0017] FIG. **9** is a plan view of the keeper shown in FIG. **7**.
- [0018] FIG. **10** is an end view of the keeper shown in FIG. **7**.
- [0019] FIG. **11** is a flat pattern for one embodiment of a cam.
- [0020] FIG. **12** is a perspective view of one embodiment of a cam.
- [0021] FIG. **13** is a side view of the keeper shown in FIG. **12**.
- [0022] FIG. **14** is a plan view of the keeper shown in FIG. **12**.
- [0023] FIG. **15** is an end view of the keeper shown in FIG. **12**.

DETAILED DESCRIPTION OF THE PRESENTLY PREFERRED EMBODIMENTS

[0024] It should be understood that the term “plurality,” as used herein, means two or more. The term “coupled” means connected to or engaged with, whether directly or indirectly, for example with an intervening member, and does not require the engagement to be fixed or permanent, although it may be fixed or permanent. The terms “first,” “second,” and so on, as used herein are not meant to be assigned to a particular component so designated, but rather are simply referring to such components in the numerical order as addressed, meaning that a component designated as “first” may later be a “second” such component, depending on the order in which it is referred. It should also be understood that designation of “first” and “second” does not necessarily mean that the two components or values so designated are different, meaning for example a first direction may be the same as a second direction, with each simply being applicable to different components. The terms “upper,” “lower,” “rear,” “front,” “fore,” “aft,” “vertical,” “horizontal,” “right,” “left,” “inboard,” “outboard” and variations or derivatives thereof, refer to the orientations of an exemplary bar lock assembly and trailer, shown in FIG. **1**, from the perspective of an operator facing the rear of the trailer.

[0025] Referring to FIG. **1**, a trailer **2** is configured with a frame **4** hingedly supporting a pair of doors **6** with two pairs of hinges **8**. The doors **6** may be rotated about axes defined by the hinges **8** between a closed position, wherein the doors **6** are closed over a doorway in the frame **4**, and an open position. A bar lock assembly **10** is secured to each door **6** and the frame **4**, and may be manipulated to secure the doors **6** to the frame **4** in the closed position, or to allow the doors **6** to be moved to the open position. It should be understood that the bar lock assemblies **10** may be used to secure other components, including other doors, and may be oriented vertically as shown in FIG. **1**, horizontally, or in other orientations.

[0026] Referring to FIGS. **1** and **2**, each bar lock assembly **10** includes at least one keeper **12** and at least one cam **14**. As shown in FIGS. **1** and **2**, each bar lock assembly **10** include a pair of keepers **12** and a pair of cams **14**, but it should be understood that a single keeper **12** and cam **14** may suffice in certain applications. The bar lock assembly **10** further includes an actuator shaft **16**, which may be rotatably secured to the door **6** with a pair of longitudinally spaced bearing plate

assemblies **18**. The bearing plate assembly **18** includes an upper bearing plate **20** having a C-shaped bearing component **24** defining a cavity and a pair of end flanges **26**, and a lower bearing plate **22** having a curved support component **28** and a pair of end flanges **30**. The upper and lower bearing plates **20**, **22** are sandwiched about the actuator shaft **16**, with the end flanges **26**, **30** being aligned and secured to the door **6** with fasteners **32**, such bolts. The actuator shaft **16** rotates in the bearing plate assemblies **18** about a rotation axis **34**. A handle **40** is rotatably secured to the actuator shaft **16** between the bearing plate assemblies **18**. The handle **40** is rotatably coupled to a clevis **42** fixedly (i.e., non-rotatably) secured to the shaft **16**, for example by welding. The clevis **42** defines a rotation axis **44** orthogonal to the rotation axis **34** of the shaft **16**. The handle **40** may be rotated about the axis **44**, but also grasped to rotate the shaft **16** about the rotation axis **34**. A locking assembly **50** includes a seal base plate **52**, which may be secured to one of the doors **6**, and a seal pivot **54** pivotally secured to the seal base plate **52** and door, for example with shaft, about a pivot axis **48**. The seal pivot **54** may be pivoted to allow a shank **56** of the handle **40** to be rotated into alignment with a slot **58** defined by the base plate **52**, and then rotated into the slot **58**, with the seal pivot **54** thereafter being pivoted over the shank **56**. A lock or other seal, such as a custom's loop, may be secured through openings in aligned flanges **60**, **62** of the seal pivot **54** and seal base plate respectively.

[0027] Referring to FIGS. 2, 3, 5 and 7-10, each keeper **12** includes a shell **80** having a base **70** and a pair of laterally spaced side walls **72**, **74** extending transversely from the base **70**. In one embodiment, the side walls **72**, **74** are orthogonal to the base **70**. Each of the side walls **72**, **74** includes or defines laterally aligned openings **76**. In one embodiment, each side wall **72**, **74** includes a pair of flanges that are longitudinally spaced apart a distance D3 to define the openings **76**. The keeper **12** further includes first and second longitudinally spaced shafts **78** extending between and connected to the side walls **72**, **74** on opposite sides of the openings **76**. In one embodiment, each shaft **78** extends through aligned openings **82**, **84** in the side walls **72**, **74**, with the shafts **78** being secured to the side walls **72**, **74**, for example by welding, or with fasteners such as nuts or other adhesives. The actuator shaft **16** is aligned with, and extends through the openings **76** in the side walls **72**, **74**. The side walls **72**, **74** are spaced apart a distance D1, and define a mouth **86** opening outwardly from the base **70**. The base **70** includes one or more mounting features **88**, shown as openings shaped to receive fasteners, such as bolts, such that the base **70** may be secured to the frame **4**. The keeper **12** is symmetrical about a center plane P, and is therefore not "handed," and may be installed at either end of the bar lock assembly **10** without needing to orient the keeper **12**.

[0028] Referring to FIG. 6, the keeper **12** may be roll formed from a flat sheet **90** of metal, having a flat pattern **92**. In one embodiment, the flat pattern **92** has an H-shape, with a center portion **94** and opposite pairs of flanges **96**, **98** that may be bent transverse to the center portion **94** by roll forming along bend lines **99**, or other known manufacturing systems, such that the flanges **96**, **98** define the side walls **72**, **74** and the keeper shell **80** has a U-shape when viewed from the end. The various openings **82**, **84**, **88** may be formed, for example by drilling, in the flat sheet **90** before forming the shell **80**. As shown in FIGS. 6 and 9, a relief **100** may be provided on opposite sides of the center portion **94**, or base **70**, to promote the forming of the side walls **72**, **74**. In another embodiment, the side walls may be continuous, with the openings **76**, **82**, **84** drilled or otherwise machined into the side walls. In various embodiments, the flat sheet **90** of metal may be made of steel, aluminum or other suitable materials.

[0029] Referring to FIGS. 2, 3, 5 and 12-15, each cam **14** includes a base **102** and a pair of laterally spaced side walls **106**, **108** extending transversely from the base **102**. Each of the side walls **106**, **108** includes or defines laterally aligned openings **110**, **112**. In one embodiment, each opening **110**, **112** is configured with a mouth **114**, **116** and an inner peripheral surface **118**, **120**, which extends less than 300 degrees around a center axis **34**, the inner periphery **118**, **120** of the openings **110**, **112** and the exterior surface of the actuator shaft **16**, which is disposed in the openings **110**, **112** and

secured thereto, for example by welding. Put another way, the openings **110**, **112** define a sector, with β defining the minor sector. In various embodiments, $\beta < 300$ degrees, and may be less than or equal to 270 degrees. The side walls **106**, **108** are spaced apart a distance **D2**, with a maximum outer, or exterior, width **W1** of the cam **14** being less than the interior width **D1** of the keeper **12**, such that the cam **14** may be nested in the keeper **12** between the side walls **72**, **74**. The side walls **106**, **108** define a mouth **124** opening outwardly from the base **102**. In this way, the U-shaped cam **14** opens toward the base **70**, and is nested between the side walls **72**, **74**, of the U-shaped keeper **12**.

[0030] The cam **14** is coupled to and rotatable with the actuator shaft **16** about the rotation axis **34** between an engaged position and a disengaged position. The cam **14** includes first and second engagement members **130**, **132** at opposite ends of each side walls **72**, **74**. The first engagement member **130** is configured with an engagement surface **134**. The engagement member **130** has an end hook defining a cavity opening towards the base **102** and away from the mouth **124**. The surface **134** has a flat portion and a curved end that is shaped to engage the shaft **78**. The second engagement member **132** is also configured with an engagement surface **136**, and includes an end hook defining a cavity opening away from the base **102**, in an opposite direction than the first cavity. In operation, as the cam **14** is rotated, the first engagement member **130** is rotated under the first shaft **78** which slides along the surface until the shaft **78** is engaged by the hook of the first engagement member **130**. At the same time, the second engagement member **132** rotates over the second shaft **78** and is engaged therewith to prevent rattling. It should be understood that the second engagement member **132**, and second shaft **78**, may be omitted in one embodiment. Even if the cam **14** is configured with only one engagement member, the cam may still be used with a keeper **12** having two shafts **78**, such that the orientation of the keeper need not be determined before installation. It should also be understood that the cam may be configured with a single side wall, having only a first engagement member, or with first and second engagement members.

[0031] In operation, the cam **14** is rotated with the actuator shaft **16** through manipulation of the handle **40** between an engaged position, wherein the first engagement member **130** is engaged with the first shaft **78** and the second engagement member **132** is engaged with the second shaft **78**, and a disengaged position, wherein the first and second engagement members **130**, **132** are disengaged with the first and second shafts **78** respectively. As shown in FIG. 5, the first engagement member **130** is disposed under the first shaft **78** and the second engagement member **132** is disposed over the second shaft **78** when the cam **14** is in the engaged position. As further referenced below, a locking device **160** may be used to secure the cam **14** to the keeper **12** to lock the bar assembly in the engaged position.

[0032] Referring to FIG. 11, the cam **14** may be roll formed from a flat sheet **140** of metal, having a flat pattern **142**. In one embodiment, the flat pattern **142** has a center portion **144** and opposite pairs of flanges **146** that may be bent transverse to the center portion by roll forming along bend lines **148**, or other known manufacturing systems, such that the flanges **146** define the side walls **106**, **108** and the cam has a U-shape when viewed from the end. In one embodiment, the side walls **106**, **108** are not bent or formed to 90 degrees with the base, but rather are angled slightly outwardly at an angle α greater than 90 degrees, for example and without limitation 92 degrees. The various openings **110**, **112**, and engagement surfaces **134**, **136** may be formed, for example by machining (e.g., laser cutting), drilling or other known manufacturing techniques, in the flat sheet **140** before forming the cam **14**. In various embodiments, the flat sheet **140** of metal may be made of steel or aluminum, or other suitable materials. Due to the unique shape of the cam **14**, and openings **110**, **112**, the actuator shaft **16** may be welded to the cam in a single welding operation, with the cam and/or welder being rotated less than 300 degrees, for example less than or equal to 270 degrees. In this way, the overall manufacturing process is greatly simplified.

[0033] Referring to FIGS. 2, 12 and 14, the cam **14** includes a lock receiving opening **150** formed in the base **102** between the first and second engagement members **130**, **132**. As shown in FIG. 2,

the lock receiving opening **150** is positioned and configured to receive a locking member **160** when the engaged position. Specifically, a shackle **162** of the lock, e.g., padlock, may be disposed through the lock receiving opening **150** and around one of the shafts **78** of the keeper **12** located proximate the lock receiving opening **150**. In this way, the bar lock assembly **10** may be secured locally in a locked configuration, without the need for the operator to access the handle **40** and locking assembly **50**, which may be elevated. A seal, or other anti-tampering device such as a zip tie, may also be secured through the receiving opening **150** and around one of the shafts **78** so as prevent and reveal any tampering.

[0034] Although the present invention has been described with reference to preferred embodiments, those skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the invention. As such, it is intended that the foregoing detailed description be regarded as illustrative rather than limiting and that it is the appended claims, including all equivalents thereof, which are intended to define the scope of the invention.

Claims

1. A bar lock assembly comprising: a keeper comprising: a base; a pair of laterally spaced side walls extending transversely from the base, wherein the side walls comprise laterally aligned openings; and first and second longitudinally spaced shafts extending between and connected to the side walls on opposite sides of the openings; an actuator shaft rotatable about a rotation axis aligned with the openings; and a cam coupled to and rotatable with the actuator shaft about the rotation axis between an engaged position and a disengaged position, wherein the cam comprises an engagement member engaged with the first shaft when rotated to the engaged position and wherein the engagement member is disengaged with the first shaft when rotated to the disengaged position.
2. The bar lock assembly of claim 1 wherein the engagement member comprises a first engagement member engaged with the first shaft when the cam is in the engaged position and disengaged from the first shaft when the cam is in the disengaged position, and further comprising a second engagement member engaged with the second shaft when the cam is rotated to the engaged position and wherein the second engagement member is disengaged with the second shaft when the cam is rotated to the disengaged position.
3. The bar lock assembly of claim 2 wherein the first engagement member is disposed under the first shaft and the second engagement member is disposed over the second shaft when the cam is in the engaged position.
4. The bar lock assembly of claim 2 wherein the cam comprises a base and a pair of a pair of laterally spaced side walls extending transversely from the base, wherein each side wall comprises the first and second engagement members.
5. The bar lock assembly of claim 4 wherein the side walls of the cam comprise laterally aligned openings, wherein the actuator shaft is disposed in the openings in the side walls of the cam.
6. The bar lock assembly of claim 5 wherein actuator shaft is welded to the side walls of the cam, wherein a weld connecting the actuator shaft and one of the side walls extends less than 300 degrees around the periphery of the actuator shaft.
7. The bar lock assembly of claim 4 wherein an exterior width of the cam between outer surfaces of the side walls of the cam is less than an interior width of the keeper between the inner surfaces of the side walls of the keeper.
8. The bar lock assembly of claim 4 wherein the cam comprises a lock receiving opening formed in the base between the first and second engagement members, wherein the lock receiving opening is positioned and configured to receive a locking member.
9. The bar lock assembly of claim 8 further comprising a lock comprising a shackle disposed through the lock receiving opening and around one of the shafts located proximate the lock

receiving member.

- 10.** The bar lock assembly of claim 2 wherein the first and second engagement members comprises first and second hooks with first and second engagement surfaces facing opposite directions.
 - 11.** A method of manufacturing the bar lock assembly of claim 1 comprising roll-forming the keeper from a piece of flat sheet metal.
 - 12.** The method of claim 11 wherein the cam comprises a U-shaped cross section, and further comprising roll-forming the cam from a piece of flat sheet metal.
 - 13.** A bar lock assembly comprising: a U-shaped keeper comprising a base, a pair of laterally spaced side walls extending transversely from the base, and a shaft extending between and connected to the side walls, and a U-shaped cam comprising a base and a pair of laterally spaced side walls extending transversely from the base, wherein the U-shaped cam opens toward and is nested between the side walls of the U-shaped keeper, wherein each side wall of the cam comprises an engagement member, wherein the cam is rotatable about a rotation axis between an engaged position wherein the engagement members are engaged with the shaft and a disengaged position wherein the engagement members are disengaged from the shaft.
 - 14.** The bar lock assembly of claim 13 wherein the shaft comprises a first shaft, and further comprising a second shaft longitudinally spaced from the first shaft, wherein the second shaft extends between and is connected to the side walls, and wherein the engagement members comprises first engagement members, and wherein each side wall of the cam further comprises a second engagement member engaged with the second shaft when the cam is rotated to the engaged position and wherein the second engagement member is disengaged with the second shaft when the cam is rotated to the disengaged position.
 - 15.** The bar lock assembly of claim 14 wherein the first engagement members are disposed under the first shaft and the second engagement members are disposed over the second shaft when the cam is in the engaged position.
 - 16.** The bar lock assembly of claim 13 wherein the side walls of the cam comprise laterally aligned openings, and further comprising an actuator shaft disposed in the openings in the side walls of the cam, wherein the actuator shaft defines the rotation axis.
 - 17.** The bar lock assembly of claim 16 wherein the actuator shaft is welded to the side walls of the cam, wherein a weld connecting the actuator shaft and one of the side walls extends less than 300 degrees around the periphery of the actuator shaft.
 - 18.** The bar lock assembly of claim 13 wherein the cam comprises a lock receiving opening formed in the base, wherein the lock receiving opening is positioned and configured to receive a locking member.
 - 19.** The bar lock assembly of claim 18 further comprising a lock comprising a shackle disposed through the lock receiving opening and around the shaft.
 - 20.** The bar lock assembly of claim 14 wherein the first and second engagement members comprises first and second hooks with first and second engagement surfaces facing opposite directions.
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